

Waterloo Engineering grad with 4 years' experience delivering robust software applications and tools.

SKILLS & EXPERIENCE

Programming Languages	C#, SQL, R, Python, C/C++, JavaScript/Typescript, Bash
Frameworks	.NET, React, XML, JSON, REST, WPF, HTML/CSS, ROS
Tools	Git, Docker, Azure DevOps, AWS, Jira, Jenkins, Visual Studio
Competencies	Numerical Modelling, Electronics, Mechanical Design, Statistics

EXPERIENCE

Abbott Point of Care <i>Intermediate Data Engineer</i>	Dec 2021 - Apr 2024 Ottawa, ON
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- Developed a .NET-based wafer-map editor to replace legacy 32-bit tooling, eliminating VM dependencies and streamlining editing across three file encoding formats. Empowered engineers with database integration, template overlays, edge exclusion tools, and undo/redo support.
- Architected a version-aware installer and update system using Squirrel, enabling seamless Dev/Val/Prod deployment and side-by-side installs - cutting update and deployment time for internal apps.
- Revamped the data engineering team's core .NET library used in Equipment Integration Apps, reducing latency and improving UX through performance tuning, bug fixes, and custom WPF control rework.
- Integrated a Filmetrics F40 spectrometer with a C++ wafer inspection GUI by wrapping vendor DLLs in a C interface and running it in a separate thread for real-time measurements and calibration.

Abbott Point of Care <i>Associate Engineer</i>	Sep 2020 - Dec 2021 Ottawa, ON
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- Met stringent deadlines leading the development, validation and deployment of three R packages which formed the backbone of a generalized ETL pipeline. Earned promotion for strong performance.
- Generated thousands of finished goods and qualification reports for various blood testing cartridges using R and Microsoft SQL Server to develop a dynamic data processing and statistical analysis backend.
- Eliminated engineering time taken for the calibration of cardiac marker blood testing products by writing a multidimensional optimization algorithm in C++ which consistently delivered best results.

Ontario Die International <i>Robotics Developer (Internship)</i>	Apr 2019 – Aug 2019 Kitchener, ON
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- Enhanced the engineering team's debugging and data retention capabilities by proposing and then building a system to record and playback robot perception, sensor data and camera footage.
- Established novel machine learning data by writing a C++ process to aggregate steel forming data.
- Improved the onboarding process by developing and documenting a bash alias repository.
- Designed real world test cases which exposed flaws in new manufacturing software features.

Avidbots Corp. <i>Software Developer (Internship)</i>	Aug 2018 – Dec 2018 Kitchener, ON
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- Enhanced the reliability of company robots by writing a Python back-up system which would recover customer setting files in the event of a failed startup.
- Built a CI/CD pipeline which detected regressions in the company's cleaning plan tools using Jenkins, Docker and Bash scripts.
- Halved the size of robot Docker images by writing a smart dependency handler to separately install compilation and runtime packages.

EDUCATION

University of Waterloo

September 2015 - April 2020

B.S. in Mechanical Engineering & Options in Management and Mechatronics

With Distinction | Dean's Honours List | U Sports Academic All-Canadian

ADDITIONAL PROJECTS

- **Hardware Accelerated Fluid Simulations** [!\[\]\(4e333a6106fc298d0ae6dff272a736ef_img.jpg\)](#) [!\[\]\(97089f8e07e24e31baa67366e358a709_img.jpg\)](#) Browser simulations built with the WebGPU API.
- **Tooner.io** [!\[\]\(9496824b8cff3a19f59b81b37b57d8b6_img.jpg\)](#) Multipage React app providing guitar players with tools to learn songs fast.
- **Interactive Travel Map** [!\[\]\(ec8d0f7e486e2280c113cd85015a8548_img.jpg\)](#) React app displaying geotagged photos on an interactive map and gallery.
- **Physics Engine** [!\[\]\(fad66fecb73aae330937d501057cafc9_img.jpg\)](#) Rigid body physics engine with no external dependencies in JavaScript.
- **Robot Simulations** [!\[\]\(a94e0943f5ecd6c1adc5223fd7677110_img.jpg\)](#) Virtual robot simulation environment using Gazebo and ROS.